

PERCENTAGE

Complete Theory, Solved Examples & Comprehensive Practice Set (English)

1. THEORY & CONCEPTS

The word **Percentage** means "per hundred". It is denoted by the symbol %. Understanding percentages is fundamental for solving various quantitative aptitude problems efficiently.

Important Conversions & Rules:

- **Fraction to Percentage:** Multiply the given fraction by 100.
 $Percentage = Fraction \times 100$
- **Percentage to Fraction:** Divide the given percentage by 100.
 $Fraction = Percentage / 100$

Key Formulas:

1. **Percentage Increase:** $Increase \% = (Difference / Original Value) \times 100$
2. **Percentage Decrease:** $Decrease \% = (Difference / Original Value) \times 100$
3. **Successive Percentage Change:** If a value undergoes successive changes of $a\%$ and $b\%$, the net effective change is:
 $Net Change \% = (a + b + (ab / 100))\%$

2. SOLVED EXAMPLES

Q1. What is 20% of 500?

Solution: $500 \times (20 / 100) = 500 \times 0.20 = 100$

Answer: 100

Q2. What percentage of 400 is 120?

Solution: $Percentage = (120 / 400) \times 100 = 120 / 4 = 30\%$

Answer: 30%

Q3. If the price of an item increases from ₹500 to ₹600, find the percentage increase in price.

Solution: Difference = $600 - 500 = 100$

Increase % = $(100 / 500) \times 100 = 20\%$

Answer: 20%

Q4. If 25% of a number is 75, find the number.

Solution: Let the number be x . $x \times (25 / 100) = 75 \Rightarrow x = (75 \times 100) / 25 = 300$

Answer: 300

Q5. A student scores 30% marks in an exam and fails by 30 marks. If the passing marks are 40%, find the maximum marks.

Solution: Percentage difference = $40\% - 30\% = 10\%$

$10\% = 30$ marks $\Rightarrow$ Maximum Marks (100%) = $(30 / 10) \times 100 = 300$

Answer: 300 marks

Q6. If the price of sugar increases by 25%, by what percentage should consumption be reduced so that expenditure remains constant?

Solution: Reduction % = $(r / (100 + r)) \times 100 = (25 / 125) \times 100 = 20\%$

Answer: 20%

Q7. If the length of a rectangle is increased by 20% and breadth is decreased by 10%, what is the net effect on its area?

Solution: Successive formula ($a = +20$, $b = -10$): $20 - 10 + ((20 \times (-10)) / 100) = 10 - 2 = +8\%$

Answer: 8% increase

Q8. A man spends 20% of his income on food, 30% on education, and 10% on transport. If he saves ₹1800, find his total income.

Solution: Total expenditure = $20\% + 30\% + 10\% = 60\%$; Savings = $100\% - 60\% = 40\%$

$40\% = ₹1800 \Rightarrow$ Total Income = $(1800 / 40) \times 100 = ₹4500$

Answer: ₹4500

Q9. A number is first increased by 20% and then decreased by 20%. Find the net percentage change.

Solution: $Loss \% = (x^2 / 100)\% = (20^2 / 100) = 400 / 100 = 4\%$

Answer: 4% decrease (Loss)

Q10. In an election between two candidates, the winning candidate gets 60% of the votes and wins by 800 votes. Find the total number of valid votes.

Solution: Difference = $60\% - 40\% = 20\%$; $20\% = 800 \Rightarrow \text{Total Votes} = (800 / 20) \times 100 = 4000$

Answer: 4000 votes

Q11. What is 20% of 25% of 300?

Solution: $300 \times (25 / 100) \times (20 / 100) = 300 \times 0.25 \times 0.20 = 15$

Answer: 15

Q12. If A's income is 25% more than B's income, by what percentage is B's income less than A's income?

Solution: Percentage less = $(r / (100 + r)) \times 100 = (25 / 125) \times 100 = 20\%$

Answer: 20%

3. PRACTICE QUESTIONS (25 Questions)

Q1. What percentage of 80 is 20?

- (a) 20% (b) 25% (c) 30% (d) 15%

Q2. If 16% of a number is subtracted from the number itself, it gives 42. Find the number.

- (a) 50 (b) 60 (c) 75 (d) 80

Q3. Convert 0.05 into a percentage.

- (a) 0.5% (b) 5% (c) 50% (d) 0.05%

Q4. 30% of 450 + 20% of 280 is equal to:

- (a) 191 (b) 185 (c) 190 (d) 195

Q5. If A's income is 25% more than B's income, then B's income is less than A's income by:

- (a) 20% (b) 25% (c) 15% (d) 10%

Q6. When a number is increased by 35%, it becomes 540. Find the original number.

- (a) 380 (b) 400 (c) 420 (d) 450

Q7. In a school, 60% of total students are boys. If the number of girls is 300, find the number of boys.

- (a) 400 (b) 450 (c) 500 (d) 600

Q8. Due to a 20% reduction in the price of sugar, a housewife can buy 4 kg more sugar for ₹240. What is the reduced price per kg?

- (a) ₹10 (b) ₹12 (c) ₹15 (d) ₹20

Q9. If the side of a square is increased by 10%, find the percentage increase in its area.

- (a) 10% (b) 20% (c) 21% (d) 19%

Q10. To pass an examination, 36% marks are required. A student gets 113 marks and fails by 85 marks. Find the maximum marks.

- (a) 500 (b) 550 (c) 600 (d) 650

Q11. What percentage of 5 kg is 20 grams?

- (a) 0.4% (b) 0.5% (c) 0.2% (d) 1%

Q12. 40% of a number is equal to 30% of 240. Find the number.

- (a) 160 (b) 180 (c) 200 (d) 220

Q13. If the price of petrol increases by 10%, by what percentage should consumption be reduced so expenditure remains unchanged?

- (a) 9.09% (b) 10% (c) 11.11% (d) 9.5%

Q14. The population of a city is 20,000. If it increases at the rate of 10% per annum, what will be the population after 2 years?

- (a) 24,000 (b) 24,200 (c) 25,000 (d) 24,500

Q15. What is 20% of 25% of 300?

- (a) 10 (b) 12 (c) 15 (d) 18

Q16. If 8% of x is equal to 4% of y , then 20% of x is equal to:

- (a) 10% of y (b) 8% of y (c) 15% of y (d) 20% of y

Q17. There are 500 eggs in a basket. 4% of them are rotten. How many good eggs are left?

- (a) 460 (b) 470 (c) 480 (d) 490

Q18. Selling an article for ₹72 results in a 10% loss. To gain 5%, what should be the selling price?

- (a) ₹80 (b) ₹84 (c) ₹88 (d) ₹90

Q19. A man spends 75% of his income. If his income increases by 20% and expenditure increases by 10%, find the percentage increase in his savings.

- (a) 50% (b) 40% (c) 60% (d) 45%

Q20. Two numbers are respectively 20% and 50% more than a third number. What percentage is the first number of the second number?

- (a) 75% (b) 80% (c) 85% (d) 90%

Q21. If 60% of $A = 30\%$ of B , then B is what percentage of A ?

- (a) 100% (b) 150% (c) 200% (d) 300%

Q22. When 60 is subtracted from a number, it reduces to 60% of itself. Find the number.

(a) 120 (b) 150 (c) 180 (d) 200

Q23. In an exam, 65% passed in History, 50% in Geography, and 25% in both. If 40 students failed in both, find the total number of students.

(a) 150 (b) 200 (c) 250 (d) 300

Q24. Due to a 10% decrease in the price of apples, a customer can buy 1 kg more apples for ₹270. Find the initial price per kg.

(a) ₹27 (b) ₹30 (c) ₹33 (d) ₹35

Q25. A vendor sold an item at a 20% profit. If he had bought it at 10% less and sold it for ₹12 less, he would have gained 30%. Find the cost price.

(a) ₹150 (b) ₹200 (c) ₹250 (d) ₹300

4. ANSWER KEY

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	(b) 25%	10	(a) 500	19	(a) 50%
2	(b) 60	11	(a) 0.4%	20	(b) 80%
3	(b) 5%	12	(b) 180	21	(c) 200%
4	(a) 191	13	(a) 9.09%	22	(b) 150
5	(a) 20%	14	(b) 24,200	23	(b) 200
6	(b) 400	15	(c) 15	24	(b) ₹30
7	(b) 450	16	(a) 10% of y	25	(b) ₹200
8	(b) ₹12	17	(c) 480		
9	(c) 21%	18	(b) ₹84		